

Analyzing Longitudinal Data With the Linear Mixed Models Procedure in SPSS

Brady T. West, MA

University of Michigan-Ann, Arbor

Many applied researchers analyzing longitudinal data share a common misconception: that specialized statistical software is necessary to fit hierarchical linear models (also known as linear mixed models [LMMs], or multilevel models) to longitudinal data sets. Although several specialized statistical software programs of high quality are available that allow researchers to fit these models to longitudinal data sets (e.g., HLM), rapid advances in general purpose statistical software packages have recently enabled analysts to fit these same models when using preferred packages that also enable other more common analyses. One of these general purpose statistical packages is SPSS, which includes a very flexible and powerful procedure for fitting LMMs to longitudinal data sets with continuous outcomes. This article aims to present readers with a practical discussion of how to analyze longitudinal data using the LMMs procedure in the SPSS statistical software package.

Keywords: *linear mixed models; multilevel models; hierarchical linear models; SPSS software*

Introduction

This article aims to present readers with a practical discussion of how to analyze longitudinal data using the linear mixed models (LMMs) procedure in the SPSS statistical software package. This extremely flexible and powerful procedure enables researchers studying individual change over time to fit a variety of advanced regression models to longitudinal data sets with

Authors' Note: Correspondence concerning this article should be addressed to Brady T. West, 283 S. Zeeb Unit 203, Ann Arbor, MI 48103; e-mail: bwest@umich.edu.

continuous dependent variables that correctly accommodate the unique statistical properties of longitudinal data.

The models discussed in this article are variously referred to in different literatures as LMMs, mixed-effects models, multilevel models, hierarchical linear models, and random coefficient models. Although they have different names, these models are all generally appropriate for studying the relationships of variables in data sets with some form of dependency introduced by a hierarchical study design. For example, in a cross-sectional educational study selecting a random sample of classrooms from within schools that were also randomly sampled, observations might be collected from students within the same classroom. In a longitudinal study, the quality of life of randomly sampled persons in an elderly population might be studied repeatedly over a period of 10 years. In both cases, there are dependencies in the observations introduced by the hierarchical study design (e.g., students within the same classroom, or repeated measurements from the same individual), and statistical models need to accommodate this dependency in the data. All of the aforementioned models are really referring to the same model: a social scientist fitting a “hierarchical linear model” to an educational data set and a physician modeling the pain of patients having a certain surgical procedure over time with a “linear mixed model” are using identical statistical tools.

Moving forward in this article, these models will be referred to as LMMs, and continuous dependent variables measured in longitudinal studies will be the focus (later sections touch on models for noncontinuous outcomes and models for clustered data sets). Many other excellent software procedures are available in general-purpose and specialized statistical software packages for fitting LMMs (e.g., SAS PROC MIXED, HLM, MLwiN), but this article focuses on the SPSS procedure. For more details, see West, Welch, and Galecki (2007).

A procedure for fitting LMMs in SPSS was first present in SPSS Version 11.0. SPSS Version 11.5 built on the original procedure by allowing for estimation of marginal means for different levels of categorical factors and additional covariance structures for the errors in the models. The version of the procedure available in SPSS 17.0, which only allows for continuous dependent variables, will be used for all examples presented in this article.

In general, LMMs provide analysts with extremely flexible statistical tools for analyses of longitudinal data sets. Some of the key advantages of LMMs relative to other techniques for the analysis of longitudinal data are highlighted below, in no particular order.

- In a repeated measures analysis of variance (rmANOVA), time-dependent (or time-varying) *continuous* covariates cannot be considered in the analysis. Only baseline values on continuous covariates can be considered. LMMs allow users to consider both time-invariant and time-varying covariates as predictors of a continuous dependent variable.
- When using rmANOVA techniques, a single missing data point on a subject's data record causes all of that subject's data to be dropped from the analysis (this is also known as listwise deletion). Under mild assumptions regarding the missing data (see West et al., 2007, section 2.9.4), LMMs are able to fully accommodate all of the data that are available for a given subject, without dropping any of the data collected for the subject. Missing data are common in longitudinal studies, especially with longer time periods of measurement, and rmANOVA techniques cannot accommodate the unbalanced data sets that may result from missing data problems.
- Several alternative covariance structures can be considered for both random errors and random effects (to be explained shortly), and models with different covariance structures can be compared using information criteria and likelihood ratio tests to determine the model with the "best fit" to a given set of longitudinal data. This flexibility is nice, but also opens up the possibility of many different models to consider. Selection of covariance structures is often driven by knowledge of subject matter and prior model-fitting investigations. One does not have the ability to consider alternative covariance structures for the data when performing rmANOVA.
- The inclusion of random effects in a LMM allows researchers to answer questions about what subject-level predictor variables (e.g., gender or treatment group) effectively explain random between-subject variance in trajectories (or growth curves) in a given longitudinal data set and to assess how much variance between subjects remains after including a certain set of fixed effects in a model. For example, one might find that gender explains 50% of the between-child variance in growth in a longitudinal study of early physical development.
- Unbalanced longitudinal data sets (data sets with different numbers of observations per subject, or subjects measured at different time points) can be easily analyzed using LMMs. Repeated measures ANOVA requires all subjects to be measured at the same time points, which represent levels of a within-subject factor (i.e., time). This is not required when fitting a LMM.

Terminology

Simple linear regression models for continuous dependent variables assume that all observations in a data set are independent of each other.

LMMs relax this assumption and allow for observations on the dependent variable to have some non-zero covariance (because they come from the same unit of analysis, or more generally from the same level of some *random factor*). A random factor is a categorical factor in a research design with levels that can be thought of as being randomly sampled from a larger population of levels. In longitudinal study designs, a “subject” variable assigning identification (ID) codes to each individual under study can be thought of as a random factor, as not all individuals in a population of interest are included in the study. Rather, one assumes that a random sample of individuals is included, and attempts to make inferences about the larger population of individuals based on the sample. Multiple observations are available in the data set for each individual, and these observations tend to be correlated with each other (because they come from the same person). LMMs allow researchers to model various forms of this correlation. Furthermore, any effects associated with the levels of a random factor enter a LMM as random variables, or *random effects*. The underlying notion here is that the values of these effects will randomly vary across study replications depending on the levels of the random factor being studied.

In contrast, *fixed factors* are categorical factors with levels that are not randomly sampled from some larger population of levels; rather, all levels are included in the study design. Common examples of fixed factors include treatment variables (where all treatments under consideration are being studied) and gender (where males and females are being studied). The levels of these variables will be the same in every theoretical replication of the current study, and any effects of these variables (or continuous predictor variables where all possible values may be measured in a study, such as height in inches) are correspondingly labeled as *fixed effects*. These are the fixed, unknown, constant regression parameters that many researchers seek to estimate when fitting standard linear regression models. The linear “mixed” model terminology comes from having a linear regression model that includes both fixed effects and random effects. This article does not aim to present a detailed theoretical discussion of LMMs, so little mathematical notation will be presented to illustrate these concepts.

Initial Data Management

Before fitting a LMM in SPSS, the data need to be in a “vertical” format, where there are several cases (or rows) corresponding to observations on an

Figure 1
SPSS (Version 17.0) Screen Shot, Indicating the Vertical Structure of the Growth Study Data Set

	subject	gender	age	distance	index	v:
1	1	F	8	21.0	1	
2	1	F	10	20.0	1	
3	1	F	12	21.5	1	
4	1	F	14	23.0	1	
5	2	F	8	21.0	2	
6	2	F	10	21.5	2	
7	2	F	12	24.0	2	
8	2	F	14	25.5	2	
9	3	F	8	20.5	3	
10	3	F	10	24.0	3	

individual unit of analysis collected over time. A researcher working with data in a “horizontal” format (where there is one row per subject and repeated measurements are contained in multiple variables) needs to use the *Restructure* command under the *Data* menu to convert the data set into a vertical structure (readers can refer to the SPSS online help for more details on this easy-to-use wizard). Here is an example of the correct data structure for a longitudinal data set, introducing the “growth study” data set that will be analyzed in this article.

Note in Figure 1 that there are four rows per subject (child), corresponding to the four observations collected over time on each child. The dependent variable, which takes on a different value in each row, is *DISTANCE*. The *AGE* variable is a time-varying predictor variable, whereas *GENDER* is a time-invariant predictor variable (taking the same value in each row for

each child). INDEX is a variable capturing the order of observation within each gender (subject 1 was the first female measured) and will not be employed for the analytic examples presented in this article because it is redundant with the “SUBJECT” variable.

The growth study data set presents an example of a balanced study design. More generally, there could be varying numbers of measurements collected across individuals under study; one child might have six observations available, whereas another might only have two. LMMs can accommodate these types of *unbalanced* study designs and data sets as well: not all individuals need to have the same number of observations and not all individuals need to be measured at the exact same time points. This flexibility is of great interest to analysts of real-world longitudinal data sets, where a perfectly balanced structure (e.g., the same number of measurements per individual, with all individuals measured at the exact same time points) is seldom seen.

Furthermore, note that the AGE variable in Figure 1 has values that are changing as a function of the measurement occasion. In contrast, values of GENDER are fixed over time for a given individual. Provided that the data are in this structure, LMMs allow researchers to analyze the relationships of both these types of independent (or predictor) variables with a continuous dependent variable of interest.

Analysis Example: The Growth Study

We now consider a step-by-step example of specifying, fitting, and interpreting a LMM in SPSS. The “growth study” data displayed in Figure 1 were collected in an orthodontic study (Potthoff & Roy, 1964), with 108 measures collected from a sample of 27 children (the children were each measured at four different ages: 8, 10, 12, and 14 years). The repeatedly measured continuous dependent variable represented the distance from the center of the pituitary gland to the pterygomaxillary fissure (in millimeters). The research was designed to study changes in this distance as a function of age for boys and girls, and the study will therefore be referred to as the “growth study” for the remainder of this example.

We first open the original growth study data set, having the “vertical” structure displayed in Figure 1, in SPSS (Version 17.0). To fit a LMM to these data, we select the SPSS menu options *Analyze* → *Mixed Models* → *Linear*. We then follow these steps, which represent one possible method of specifying a LMM in SPSS:

1. The first dialog box asks the user to identify *Subjects* variables and *Repeated* variables. The “Subjects” variables are the variables that identify blocks of repeated measures on subjects, which one would expect to have some non-zero correlation (i.e., several observations of growth collected over time on the same child). The “Repeated” variables index the repeated measurements in the vertical data set (if applicable). One does not have to define any variables at this point, and can simply click the Continue button. In the growth study data set, there are four measures (rows) for each subject, so we identify SUBJECT as a “Subjects” variable, and AGE as a “Repeated” variable indexing the repeated measures on each subject.
2. After selecting the variables that define the repeated measures (AGE), SPSS allows for the selection of a *Repeated Covariance Type*. In standard linear regression models, *all* random errors associated with observed values of the dependent variable are assumed to be independent (having zero covariance) and have constant variance (this is one type of *covariance structure*). The covariance structure for the random errors can be generalized in LMMs, which is important, because longitudinal studies will generally introduce a non-zero covariance between random errors associated with two measurements on the same individual. When there are repeated measures collected over time on a subject, the errors are no longer assumed to be completely independent of each other, and SPSS allows one to estimate the covariance structure of the random errors. For a better idea of what the possible structures look like for a set of repeated observations on an individual, one can click Help, and then click the *Covariance Structures* link.

There are many covariance structures to choose from in this SPSS procedure. In the Growth Study data set, we assume that adjacent observations on the same SUBJECT (indexed by AGE) will have errors with a higher correlation than observations that are farther apart, so we specify a *first-order autoregressive* [*AR(1)*] covariance structure, and click Continue. Errors associated with observations for different levels of SUBJECT (i.e., different subjects) are assumed to be uncorrelated. This procedure allows analysts to try different covariance structures and determine which one provides the best fit to the longitudinal data being analyzed (to be discussed more later).

3. After clicking Continue, one defines the continuous *Dependent Variable* in the model, any categorical *Factors* under consideration as predictors and any continuous (or interval) *Covariates* under consideration as predictors. These variables include both time-varying and time-invariant

predictors. In the growth study, we define the dependent variable as DISTANCE, the categorical factor as GENDER, and the continuous covariate as AGE. By specifying AGE as a continuous covariate, we indicate that we wish to estimate the functional relationship of AGE with DISTANCE. (An alternative would be to specify AGE as a factor, and estimate means for DISTANCE at each level of age, rather than fitting a line or a curve.) Variables containing weights for the cases can be identified in the *Residual Weight* box (this is beyond the scope of this article).

4. After selecting the variables that will be involved in fitting the model, one can click the *Fixed...* button to define the *fixed effects* that will be included in the model. Recall that the fixed effects are the constant (and unknown) regression coefficients (or regression parameters) associated with either the continuous covariates or the categorical factors, where all possible theoretical values of the variables (in terms of the desired inferences based on the model) are present in the data set (e.g., GENDER). At this point, one can also include the fixed effects of interactions that one would like to see in the model. In the growth study, we include fixed effects associated with AGE, GENDER, and the interaction between AGE and GENDER (specified by holding down the CTRL button and clicking on both variables). The fixed effects can be added to the model using the Add button.¹
5. When done adding fixed effects of interest to the model, click Continue, and then click the *Random...* button to define the *random effects* that will be included in the model. Random effects will be associated with levels of the *Subjects* variables selected earlier, where the categories of the “subjects” variables represent a random sample of categories from some theoretical population of interest (e.g., all children in the growth study). Random effects might help define the subject-specific intercepts in a model, or the subject-specific coefficients of predictors (like AGE), allowing different subjects to have *unique trajectories*. When there are random effects included in a model, SPSS can estimate the *variances* of the random effects (in addition to their *covariances*), allowing one to investigate the amount of between-subject variance in terms of the coefficients in your model.

In the growth study, we consider the possibility that different subjects might have different intercepts² defining their growth trajectories, in addition to different slopes (representing the relationship of age with growth). To specify these two random effects, make sure that the SUBJECT variable is added to the *Combinations* window, which appears at the bottom of the window that is currently open entitled “Linear Mixed Models: Random Effects.” This step indicates that the random effects will be associated with

the different levels of the SUBJECT variable. Then, check “Include Intercept” to include a random effect associated with the intercept for each Subject, and add AGE to the *Model* window to add a random effect associated with the AGE coefficient (i.e., slope) for each subject. Define the *Covariance Type* as “Unstructured,” which in short allows for the estimation of the variances of the random effects, in addition to their *covariance* (for example, a child with a higher random intercept, representing higher initial growth, may have a smaller increase in growth over time, or a growth curve that is not as steep: this would define a negative covariance between the two random effects). When finished, click Continue.

6. Next, the *Estimation . . .* button provides users with some technical choices for the model fitting. For most analyses, the default choices will do. We estimate the parameters in the Growth Study model (the fixed effects and the variances and covariance of the two random effects) using a mathematical technique known as restricted maximum likelihood (REML) estimation, which is the default, and tends to result in unbiased estimates of the variances and covariances (as opposed to maximum likelihood, or ML, estimation; more discussion of this issue follows below).
7. The *Statistics . . .* button provides users with choices for which types of statistical output they would like to see. Reasonable choices for most analyses include *Parameter Estimates* and *Tests for Covariance Parameters*, which will display estimated regression coefficients (or estimated fixed effects) as well as estimated variance and covariance parameters for the random effects or repeated measures that are specified for a particular model.
8. The *EM Means . . .* button allows users to display estimated marginal means (or predicted values of the dependent variable, fixed at the means of any covariates in the model) on the dependent variable for the different levels of the fixed factors and also compare them statistically. In the Growth Study, we display EM Means for the levels of the categorical fixed factor GENDER, and compare the means by checking the *Compare main effects* option. Note that no confidence interval adjustment for multiple comparisons is required because we are only comparing two groups. Click Continue when finished.
9. Finally, users can (and should) also *Save . . .* residuals and predicted values for performing the standard diagnostics associated with regression modeling (e.g., assessing assumptions of normality and constant variance for the random errors of the specified model for each value of age). This article does not discuss diagnostics for LMMs in detail, but interested readers can refer to West et al. (2007) for more details.

When finished setting up a LMM analysis using the SPSS menus as described above, users are recommended to click *Paste* to save the SPSS syntax for the analysis. This simplifies the specification and estimation of future models and allows users to keep an electronic summary of the model fitted using SPSS syntax. Here is the syntax that corresponds to the analysis specified above:

```
MIXED
distance BY gender WITH age
/CRITERIA = CIN(95) MXITER(100) MXSTEP(5) SCORING(1)
SINGULAR(0.000000000001) HCONVERGE(0, ABSOLUTE) LCON-
VERGE(0, ABSOLUTE)
PCONVERGE(0.000001, ABSOLUTE)
/FIXED = age gender age*gender | SSTYPE(3)
/METHOD = REML
/PRINT = SOLUTION TESTCOV
/RANDOM INTERCEPT age | SUBJECT(subject) COVTYPE(UN)
/REPEATED = age | SUBJECT(subject) COVTYPE(AR1)
/SAVE = PRED RESID
/EMMEANS = TABLES (gender) COMPARE ADJ(LSD).
```

What follows is a sampling of the SPSS output that results from running the syntax above on the growth study data set, in addition to a discussion of what each set of output represents.

Analysis Example: Results

The *Model Dimension* table (Table 1) details the parameters in the LMM that will be estimated by the procedure, and the number of “subjects” in the analysis based on the random effects included in the model (27 in this case, corresponding to 27 children). Users are strongly recommended to identify what each parameter in this table represents before interpreting the resulting estimates. Note that nine parameters are being estimated in the current model: four regression parameters (the fixed effects, or the intercept, the linear effect of age, the effect of gender, and the interaction between age and gender), three variance-covariance parameters associated with the two random effects (two variances and one covariance), and two variance-covariance parameters associated with the random errors (the variance of the errors, and the correlation of two adjacent errors for the same subject).

Table 1
The Model Dimension Output

Model Dimension ^a		Number of Levels	Covariance Structure	Number of Parameters	Subject Variables	Number of Subjects
Fixed Effects	Intercept	1		1		
	Age	1		1		
	Gender	2		1		
	Gender × age	2		1		
Random Effects	Intercept + age ^b	2	Unstructured	3	Subject	
Repeated Effects	Age		First-Order			
		4	Autoregressive	2	Subject	27
Total		12		9		

^a Dependent Variable: distance Distance (mm) from center of pituitary to pteryo-maxillary fissure.

^b As of version 11.5, the syntax rules for the RANDOM subcommand have changed. Your command syntax may yield results that differ from those produced by prior versions. If you are using SPSS 11 syntax, please consult the current syntax reference guide for more information.

Table 2
The Information Criteria Output

Information Criteria ^a	
−2 restricted log-likelihood	428.808
Akaike's information criterion (AIC)	438.808
Hurvich and Tsai's criterion (AICC)	439.420
Bozdogan's criterion (CAIC)	457.030
Schwarz's Bayesian criterion (BIC)	452.030

The information criteria are displayed in smaller-is-better forms.

^a Dependent Variable: distance Distance (mm) from center of pituitary to pteryo-maxillary fissure.

The *Information Criteria* table (Table 2) provides useful model fit criteria for comparing alternative models. These values are useless in the context of a single model, but are extremely useful when comparing the fits of

Table 3
Type III Tests of Fixed Effects

Type III Tests of Fixed Effects ^a				
Source	Numerator <i>df</i>	Denominator <i>df</i>	<i>F</i>	Sig.
Intercept	1	24.984	460.530	.000
Age	1	24.655	87.180	.000
Gender	1	24.984	0.654	.426
Gender × age	1	24.655	5.581	.026

^a Dependent variable: distance Distance (mm) from center of pituitary to pteryo-maxillary fissure.

competing models, and especially nested models (where one model contains a subset of the parameters in a competing model). *Likelihood ratio tests* for certain parameters are performed by using the -2 log-likelihood values in this table, and we discuss this process in more detail in an upcoming section. In general, the smaller the values of these criteria, the better the fit of the model.

The *Type III Tests of Fixed Effects* (Table 3) represent overall (or omnibus) tests of significance for the predictor variables included in the model, taking the other predictors in the model into account. These F tests are especially useful to investigate when looking at omnibus tests of significance for predictors in the model that involve several fixed effects, such as complex interactions between predictors (like the interaction between gender and age). Note that the denominator degrees of freedom used for the F statistics are not whole numbers; this is due to the fact that SPSS is using the Satterthwaite approximation method to compute the denominator degrees of freedom when fitting LMMs. When fitting LMMs with explicit random effects and correlated errors, these F statistics no longer follow exact F distributions, and degrees of freedom are often approximated. Currently, the LMMs procedure in SPSS only provides users with the Satterthwaite approximation method.

For an example interpretation, the Type III F test for the interaction between age and gender is significant at the 5% level, suggesting that the relationship of age with growth varies depending on gender [$F(1, 24.655) = 5.581, p = 0.026$].

The *Estimates of Fixed Effects* (Table 4) represent the maximum likelihood estimates of the *fixed effect parameters* (or regression coefficients) in the LMM being fitted. Readers should review this output carefully, because these coefficients represent the “meat” of the analysis results. There is a significant positive linear relationship of AGE with DISTANCE for males

Table 4
Estimates of Fixed Effects

Estimates of Fixed Effects ^a							
Parameter	Estimate	SE	df	t	p	95% Confidence Interval	
						Lower Bound	Upper Bound
Intercept	16.152436	0.998461	24.984	16.177	.000	14.096000	18.208871
Age	0.797950	0.087068	24.655	9.165	.000	0.618503	0.977396
[Gender = F]	1.264698	1.564289	24.984	.808	.426	-1.957117	4.486513
[Gender = M]	0 ^b	0					
[Gender = F] × age	-0.322243	0.136409	24.655	-2.362	.026	-0.603382	-0.041105
[Gender = M] × age	0 ^a	0					

^a Dependent Variable: distance Distance (mm) from center of pituitary to pterygo-maxillary fissure.

^b This parameter is set to zero because it is redundant.

(the reference level of GENDER), with the estimated fixed effect being 0.798. This suggests that for males, a 1-year increase in age results in an increase of 0.798 in mean distance (or growth), and this fixed effect is significantly different from 0 based on the *t* test (which also uses a Satterthwaite approximation of the degrees of freedom for the *t* statistic). There is also a significant interaction between GENDER and AGE, with the slope decreasing by 0.322 for females (meaning that the net slope for females is 0.476). This change in the relationship is significant, based on the estimated fixed effect ($p = 0.026$). The estimated fixed effect for GENDER represents the change in the intercepts (or predicted values when AGE = 0) for females compared to males. So we have evidence of a significant interaction between AGE and GENDER.

Readers should note that the significance values for the tests of the *individual* fixed effects will not always correspond exactly with the significance values for the *omnibus* tests of the fixed effects (which may involve more parameters).

The *Estimates of Covariance Parameters* (Table 5) represent the maximum likelihood estimates of the *covariance parameters* (variances of the random effects, covariances of the random effects, variances and covariances of the random errors, etc.) in the LMM that is being fitted. There are two covariance parameters defining the error covariance structure in the

Table 5
Estimates of Covariance Parameters

Estimates of Covariance Parameters ^a							
Parameter		Estimate	SE	Wald Z	Sig.	95% Confidence Interval	
						Lower Bound	Upper Bound
Repeated Measures	AR1	1.192410	0.235547	5.062	.000	0.809619	1.756188
	AR1 rho	-0.473280	0.187112	-2.529	.011	-0.756028	-0.041674
Intercept + age [subject = subject]	UN (1,1)	11.377451	4.928971	2.308	.021	4.867265	26.595299
	UN (2,1)	-0.814984	0.415695	-1.961	.050	-1.629731	-0.000236
	UN (2,2)	0.084546	0.039080	2.163	.031	0.034169	0.209193

^a Dependent Variable: distance Distance (mm) from center of pituitary to pteryo-maxillary fissure.

model for the growth study data: the variance of the random errors [AR1 diagonal in the AR(1) covariance structure], and the correlation of adjacent errors (in terms of the amount of time between adjacent measures, or 2 years; this is AR1 rho). The variance of the random errors is estimated to be 1.19, which means that random errors associated with the growth observations (taking the fixed effects into account) are sampled from a normal distribution with variance 1.19. The estimates suggest that adjacent errors in the model actually have a *negative* correlation (−0.473), which means that when conditioning on the fixed effects and random effects in this model, errors associated with observations from the same child at adjacent time points have a negative correlation; this might bring the covariance structure into question, prompting investigation of alternative covariance structures.

The next three covariance parameters are associated with the two random effects that are included in this model. These covariance parameters define a 2 × 2 variance–covariance matrix for the random effects, illustrated below:

$$\begin{bmatrix} \text{Var}(\text{random intercepts}) & \text{Cov}(\text{random intercepts, random slopes}) \\ \text{Cov}(\text{random intercepts, random slopes}) & \text{Var}(\text{random slopes}) \end{bmatrix} \\ = \begin{bmatrix} \text{UN}(1, 1) & \text{UN}(2, 1) \\ \text{UN}(2, 1) & \text{UN}(2, 2) \end{bmatrix}$$

The first parameter, represented by UN(1,1) (corresponding to the element in the first row and first column of the unstructured 2×2 variance-covariance matrix for the two random effects), represents the estimated variance of the random subject effects associated with the *intercept* in the model (each child is allowed to have a unique intercept, or predicted value when AGE = 0, when these random effects are included²). There is initial evidence of significant variance in these random effects based on the approximate Wald test presented by SPSS ($p = .02$), suggesting that the intercepts do in fact vary from child to child (a likelihood ratio test would be a more appropriate test than this Wald test; more details on this will be presented shortly). Readers should note that the estimated variance of the random intercepts will change depending on how the AGE variable is centered; in this example, the UN(1,1) estimate corresponds to the estimated between-child variance of the intercepts when the AGE variable is equal to 0, which may not be entirely useful for researchers.

The next covariance parameter, represented by UN(2,1) (corresponding to the element in the second row and first column of the random effects variance-covariance matrix), is the estimated *covariance* of the two random effects. The estimated covariance is negative, suggesting that the higher the random effect for the intercept (the larger a subject's intercept), the smaller the random effect for the age slope (the smaller the subject's slope). This finding is quite common in longitudinal studies and makes sense in this context: the more distance that a child has to begin with, the less growing that they have to do.

The last parameter, represented by UN(2,2), is the estimated variance of the random subject effects associated with the *age slope* in the model. There is also evidence of significant variance in these random effects ($p = .03$), which suggests that the slopes tend to vary from child to child. SPSS users can visualize this variance by investigating a multiple line graph, where each child has a separate line plotted (*Graphs* → *Legacy Dialogs* → *Line* → *Multiple*, Category Axis = AGE, Other statistic = MEAN(DISTANCE), Define Lines by = SUBJECT).

The *Estimated Marginal Means* section of the output (Tables 6 and 7) presents estimates of the marginal means for the dependent variable (or predicted values based on the estimated fixed effects in the model) for different groups defined by some categorical factor. Note that the estimates for boys and girls are evaluated at the mean of the continuous covariate in the model (AGE). As requested in the syntax, SPSS also produces pairwise comparisons of the estimated marginal means, allowing analysts to test hypotheses about the means for selected values of the covariates. In this example, the

Table 6
Estimated Marginal Means

Gender	Mean	SE	df	Estimates ^a	
				95% Confidence Interval	
				Lower Bound	Upper Bound
Girl	22.650 ^b	.588	24.997	21.439	23.861
Boy	24.930 ^b	.488	24.997	23.926	25.934

^a Dependent Variable: distance Distance (mm) from center of pituitary to pteryo-maxillary fissure.

^b Covariates appearing in the model are evaluated at the following values: age Age in years = 11.00.

Table 7
Pairwise Comparisons of Estimated Marginal Means

(I) Gender	(J) Gender	Mean Difference (I-J)	SE	df	<i>p</i> ^b	Pairwise Comparisons ^a	
						95% Confidence Interval for Difference ⁸	
						Lower Bound	Upper Bound
Girl	Boy	-2.280*	.764	24.997	.006	-3.853	-0.706
Boy	Girl	2.280*	.764	24.997	.006	0.706	3.853

Based on estimated marginal means,

* The mean difference is significant at the .05 level.

^a Dependent Variable: distance Distance (mm) from center of pituitary to pteryo-maxillary fissure.

^b Adjustment for multiple comparisons: least significant difference (equivalent to no adjustments).

difference in means between boys and girls at the mean age is significant at the .01 level ($p = .006$).

This section has provided a very basic overview of the output that analysts fitting LMMs in SPSS will see and is designed to familiarize analysts with the output tables. Specific interpretations will vary for different analyses.

Likelihood Ratio Tests for Covariance Parameters

When reporting the estimates of the covariance parameters (Table 5 in the last section), SPSS also reports Wald tests of the null hypotheses that each parameter is equal to 0 (provided that the appropriate TESTCOV option is present in the /PRINT subcommand of the MIXED syntax). These Wald tests generally have unfavorable statistical properties, and likelihood ratio tests are more appropriate tests for covariance parameters (assuming that sample sizes are fairly large). In general, when testing null hypotheses involving covariance parameters that represent variances of random effects and the null hypothesis states that a given variance parameter is equal to zero, *specialized* likelihood ratio tests should be performed. We will return to this idea shortly. First, classical likelihood ratio tests comparing *nested* models (where one model only has a subset of parameters in a larger model; note that this idea differs from that of nested effects, introduced earlier) are generally computed as follows:

Model 1 (reduced model): X_1 parameters, $-2 \text{ REML LL} = \text{LL1}$

Model 2 (larger model): $X_1 + X_2$ parameters, $-2 \text{ REML LL} = \text{LL2}$

The difference in $-2 \text{ REML log-likelihood}$ values between the two models, or $\text{LL1} - \text{LL2}$, asymptotically (i.e., assuming large samples) follows a χ^2 distribution with X_2 degrees of freedom (where X_2 is the number of additional parameters in the larger model, as defined above). This is where the information criteria reported in the SPSS output comes in handy, and this result proves to be quite useful for testing a wide variety of hypotheses.

Suppose, for example, that we wanted to test whether the AR(1) covariance structure was appropriate for the random errors associated with observations for a given child in the model specified above. From Table 2, the -2 REML LL value for this model (the “large” model) is 428.81. We refit the model without the AR(1) covariance structure using the following syntax:

```
MIXED
distance BY gender WITH age
/CRITERIA = CIN(95) MXITER(100) MXSTEP(5) SCORING(1)
SINGULAR(0.000000000001) HCONVERGE(0, ABSOLUTE) LCON-
VERGE(0, ABSOLUTE)
PCONVERGE(0.000001, ABSOLUTE)
/FIXED = age gender age*gender | SSTYPE(3)
/METHOD = REML
```

```

/PRINT = SOLUTION TESTCOV
/RANDOM INTERCEPT age | SUBJECT(subject) COVTYPE(UN)
/SAVE = PRED RESID
/EMMEANS = TABLES(gender) COMPARE ADJ(LSD).

```

Note that the /REPEATED subcommand has been removed. This can also be done via the menus by not specifying a “Repeated” variable in the very first screen. By doing this, we are indicating that the errors do not have any type of specialized covariance structure and are simply independent and identically distributed with homogeneous variance (as in a standard linear regression model). Table 8 presents the information criteria for the refitted model, generated by running the new syntax above.

We wish to test the null hypothesis that the AR(1) rho parameter is equal to 0 (which means that there is no correlation between adjacent errors associated with observations on the same individual). We can do so by computing the difference in the -2 REML LL values between the two models:

$$432.58 - 428.81 = 3.77$$

Based on likelihood ratio theory, this difference represents a test statistic which follows a χ^2 distribution with 1 degree of freedom (because our new reduced model has one fewer parameter or the AR(1) rho parameter). The p value for this χ^2 statistic (3.77) is roughly .05, which suggests that there is a significant difference between the -2 REML LL values and that the null hypothesis of zero correlation between adjacent errors should be rejected. Think of it this way: a model with a higher LL will have a “better” fit to the data, so if the LL becomes significantly “worse” when dropping a parameter, we should keep that parameter in the model. Recall that the negative estimated covariance between adjacent errors was a bit strange and may require further investigation.

This was a general example for covariance parameters. When testing covariance parameters that represent variances of random effects, a slight modification to the test is necessary. Suppose, for example, that we wanted to test the null hypothesis that the variance of the random age slopes for the children [UN(2,2) in the SPSS output] was equal to 0. This is a variance parameter, and technically a variance parameter cannot be equal to 0 (every child would need to have the exact same slope). The fact that the null hypothesis specifies a parameter *on the boundary of the parameter space* brings up theoretical problems with the simple likelihood ratio theory above (see Verbeke & Molenberghs, 2000, or West et al., 2007, for more details). As a result, the likelihood ratio test becomes a bit more complicated. In the

Table 8
**Information Criteria for the Reduced Model Without the AR(1)
 Covariance Structure**

Information Criteria^a

–2 Restricted log-likelihood	432.582
Akaike's information criterion (AIC)	440.582
Hurvich and Tsai's criterion (AICC)	440.986
Bozdogan's criterion (CAIC)	455.159
Schwarz's Bayesian criterion (BIC)	451.159

The information criteria are displayed in smaller-is-better forms.

^a Dependent Variable: distance Distance (mm) from center of pituitary to pteryo-maxillary fissure.

case of a model with random intercepts and random slopes (two variances and a covariance associated with the random effects, assuming an unstructured covariance structure for the random effects), if one of the variances is dropped from the model, the difference in –2 REML LL values between the two models follows a *mixture* of χ^2 distributions, with 1 and 2 degrees of freedom, and equal weight 0.5.

Consider the analysis above. In the full model, the –2 REML LL is equal to 428.81 (Table 2). We drop the random age slopes using the following syntax (note how AGE is dropped from the /RANDOM subcommand):

```
MIXED
distance BY gender WITH age
/CRITERIA = CIN(95) MXITER(100) MXSTEP(5) SCORING(1)
SINGULAR(0.000000000001) HCONVERGE(0, ABSOLUTE) LCON-
VERGE(0, ABSOLUTE)
PCONVERGE(0.000001, ABSOLUTE)
/FIXED = age gender age*gender | SSTYPE(3)
/METHOD = REML
/PRINT = SOLUTION TESTCOV
/RANDOM INTERCEPT | SUBJECT(subject) COVTYPE(UN)
/REPEATED = age | SUBJECT(subject) COVTYPE(AR1)
/SAVE = PRED RESID
/EMMEANS = TABLES(gender) COMPARE ADJ(LSD).
```

Table 9 presents the information criteria from the reduced model.

Keep in mind that the reduced model has *two* fewer parameters (since we also necessarily drop the *covariance* of the two random effects as well as the

Table 9
Information Criteria From the Reduced Model Without the
Random Age Effects

Information Criteria ^a	
–2 Restricted log-likelihood	433.708
Akaike’s information criterion (AIC)	439.708
Hurvich and Tsai’s criterion (AICC)	439.948
Bozdogan’s criterion (CAIC)	450.641
Schwarz’s Bayesian criterion (BIC)	447.641

The information criteria are displayed in smaller-is-better forms.

^a Dependent Variable: distance Distance (mm) from center of pituitary to pteryo-maxillary fissure.

variance of the random age slopes; there is only one random effect now, so no covariance between two random effects can be estimated). Based on the result described above, the difference in –2 REML LL values ($433.71 - 428.81 = 4.90$) follows a mixture of χ^2 distributions with 1 and 2 degrees of freedom and equal weights of 0.5. The p value for testing the null hypothesis that the variance of the random slopes is therefore computed as follows:

$$\begin{aligned} p \text{ value} &= 0.5 \times P(\chi^2(1) > 4.9) + 0.5 \times P(\chi^2(2) > 4.9) \\ &= 0.5 \times 0.027 + 0.5 \times 0.086 = 0.057. \end{aligned}$$

The p value falls above the standard .05 significance level, so we do not have very strong evidence in favor of keeping the random slopes in the model. This is a marginal result, so we could really go either way in deciding whether to include the random slopes. There is some evidence of variation between the children in terms of the random age slopes. When in doubt, it does not hurt to keep these parameters in the model and discuss the marginal significance.

In general, when fitting LMMs, it is a good idea to first fit a model with all fixed effects of interest (including interactions) and establish an appropriate covariance structure for the random effects and errors using the types of tests described above. Once an appropriate covariance structure has been determined, tests for the fixed effects can be considered, and one can reduce the fixed effects in the model. This is one possible strategy; see West et al. (2007) or Verbeke and Molenberghs (2000) for more details.

Mixed Models for Nonnormal Outcomes

Unfortunately, the current release of SPSS (Version 17.0) does not allow users to fit *generalized* LMMs (GLMMs) to dependent variables that are nonnormal (e.g., binary outcomes, categorical outcomes, count outcomes, etc.). Other current software options for these types of models include the HLM software, SAS PROC GLIMMIX, SAS PROC NL MIXED, and the *xtmelogit* and *xtmepoisson* commands in Stata (Version 10). Development of software for fitting these types of models has accelerated, and this feature will hopefully be available in newer releases of SPSS.

Analyzing Clustered Longitudinal Data

This article has presented an example analysis of a simple longitudinal data set, where repeated measures were collected from sampled individuals. Some study designs may involve sampling clusters of individuals, and then following selected individuals within the clusters over some period of time. These types of study designs have the potential to introduce multiple levels of correlation in the observations collected: observations within the same cluster (e.g., a business) may have some general non-zero correlation, and observations within the same individual within a cluster (e.g., an employee followed over time) may have a different correlation structure. The LMMs procedure in SPSS can also accommodate these types of advanced correlation structures, but some care is required in specifying the appropriate combination of *Subjects* and *Repeated* variables when setting up the analysis. Interested readers can refer to West et al. (2007, chapter 7) for an example of using SPSS for this type of analysis.

Summary

This article aimed to provide readers with a practical discussion of how to use the SPSS software to fit LMMs to longitudinal data sets with continuous dependent variables. LMMs are extremely flexible statistical tools for analysts of longitudinal data, and the LMMs procedure in SPSS enables analysts to easily fit a wide range of sophisticated statistical models to longitudinal data that can be used to address a variety of research hypotheses about subjects in the population from which the data were collected. A

failure to correctly model all the unique features of longitudinal data (e.g., correlation of random errors associated with adjacent observations, or random effects capturing between-subject variance) when testing research hypotheses can lead to erroneous research conclusions, and researchers should use this type of software to take full advantage of the information contained within longitudinal data sets.

Notes

1. If levels of fixed factors are nested by design, one can specify nested fixed effects by clicking the radio option for “Build nested terms,” selecting the fixed factor with levels nested within a second fixed factor, clicking the “down arrow” button underneath the list of factors and covariates, clicking the (Within) button, selecting the second factor, again clicking the “down arrow” button, and then clicking Add.

2. An alternative analysis approach might involve centering the AGE variable at some value (e.g., 10), so that the intercept in the model can be interpreted as an expected value of the dependent variable at a given age, such as age 10. This could be accomplished by subtracting the value in question (e.g., 10) from all values of AGE, and then using the “centered” variable as a continuous predictor in the model. Readers should note that without centering the age variable, the intercept is interpreted as the expected value of the dependent variable at age 0. For more information on centering, readers can refer to Paccagnella (2006).

References

- Paccagnella, G. (2006). Centering or not centering in multilevel models: The role of the group mean and the assessment of group effects. *Evaluation Review*, 30, 66.
- Potthoff, R. F., & Roy, S. N. (1964). A generalized multivariate analysis of variance model useful especially for growth curve problems. *Biometrics*, 51, 313-326.
- Verbeke, G., & Molenberghs, G. (2000). *Linear mixed models for longitudinal data*. Berlin, Germany: Springer-Verlag.
- West, B. T., & Welch, K. B., & Galecki, A. T. (2007). *Linear mixed models: A practical guide using statistical software*. Florida: Chapman Hall/CRC Press.